

# MADHAVAN NAGARAJAN

Register Number: 2403717610421101

## COIMBATORE INSTITUTE OF TECHNOLOGY

### Smart Queue & Slot Booking System

#### Problem Statement: Smart Queue & Slot Booking System

In many service-based environments such as college offices, laboratories, clinics, canteens, and help desks, people spend a large amount of time waiting in physical queues. Manual queue systems often create confusion because users do not know their current token position, approximate waiting time, or whether the service counter is active or delayed.

The problem lies in the absence of a transparent and organized digital queue management system. Users are forced to physically wait without proper information, while administrators face difficulty in managing tokens, skipped users, cancellations, service progress, and crowd levels. This leads to time wastage, overcrowding, poor user experience, and inefficient service handling.

#### Proposed Solution: Smart Queue & Slot Booking System

We propose building a web-based Smart Queue & Slot Booking System using the MERN stack. The system allows users to book a token or time slot online, track their queue status, and view a simple live waiting indicator. The admin can manage the queue by calling the next user, updating token status, skipping unavailable users, and monitoring the crowd level.

The system will:

- Allow users to register, log in, and book a token or slot for a selected service.
- Generate a unique token number for each booking.
- Show the current serving token, user queue position, and estimated waiting time.
- Allow users to cancel their token if they are unable to attend.
- Provide an admin dashboard to call, complete, skip, or cancel tokens.
- Display a simple crowd level indicator such as Low, Medium, or High.
- Improve transparency and reduce unnecessary physical waiting time.

This solution will help institutions and service counters manage queues efficiently, reduce crowding, save user time, and provide a more organized service experience.

#### Main Modules

- Authentication Module - User registration, user login, admin login, and role-based access.
- Service Selection Module - Users can select the required service such as office work, lab submission, canteen, clinic, or help desk.
- Token Booking Module - Generates and stores token details with date, time, service type, and status.
- Queue Tracking Module - Shows live queue position, current token, and estimated waiting time.
- Admin Dashboard Module - Admin can view active tokens and update queue status.
- Crowd Level Module - Displays crowd status based on the number of waiting users.

- Report Module - Shows total tokens, completed tokens, skipped tokens, cancelled tokens, and average waiting time.

## **Frontend Tools**

- React.js - Frontend user interface framework.
- Tailwind CSS / Material UI - Styling and responsive design.
- React Router - Navigation between user pages and admin dashboard.
- Axios - Connecting frontend with backend APIs.
- Redux Toolkit / Context API - State management for login, token, and queue status.

## **Backend Tools**

- Node.js - Runtime environment for backend development.
- Express.js - Backend framework for creating REST APIs.
- MongoDB - Database for storing users, services, tokens, and queue status.
- Mongoose - Object Data Modeling library for MongoDB.
- JWT Authentication - Secure login and role-based authorization.
- bcrypt.js - Password hashing for secure user credentials.

## **APIs & Services**

- Socket.io - Optional real-time queue updates without page refresh.
- QR Code Generator - Optional QR-based token verification.
- Render / Vercel / Netlify - Frontend deployment.
- Render / Railway / AWS - Backend deployment.
- MongoDB Atlas - Cloud database hosting.

## **Database Collections**

- Users Collection - Stores user name, email, password, and role.
- Services Collection - Stores service name, description, average service time, and active status.
- Tokens Collection - Stores token number, user ID, service ID, status, booking time, called time, and completion time.
- Queue Status Collection - Stores current token, total waiting users, and crowd level for each service.

## **Advantages**

- Reduces time wasted in physical queues.
- Provides transparent queue position and waiting information.
- Helps admins manage users in an organized way.
- Avoids overcrowding near counters.
- Easy to build and demonstrate using the MERN stack.
- Covers important web development concepts such as login, CRUD, dashboard, API integration, and status updates.
- Can be extended with real-time updates, QR token, analytics, and multi-counter support.

## **Limitations**

- The system depends on proper admin usage to maintain queue accuracy.
- Estimated waiting time may vary based on actual service speed.

- Basic version may not provide true real-time updates unless Socket.io is implemented.
- Internet connectivity is required for users and admins.
- If users book tokens but do not arrive, the queue flow may be affected.

### **Expected Outcome**

The final project will be a functional MERN web application where users can book tokens, monitor queue progress, and receive clear status updates. The admin will be able to manage the queue through a dedicated dashboard. The project will demonstrate practical full-stack development skills and solve a real-world problem faced in colleges, clinics, offices, and service counters.